

Experiment 5

Aim: To demonstrate the performance of a linear regression by using chosen database with python code

Program

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import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

iris = load_iris()
data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target
print(data.head())

X = data[['sepal length (cm)']]
y = data['sepal width (cm)']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
print(f'Mean Squared Error on test set: {mse:.2f}')

plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, color='r', marker='*', label='Actual')
plt.plot(X_test, y_pred, color='y', linewidth=3, label='Predicted')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Linear Regression: Sepal Width vs Sepal Length')
plt.legend()
plt.show()
```

```

new_sample = pd.DataFrame([[5]], columns=['sepal length (cm)'])
predicted_width = model.predict(new_sample)
print(f'The predicted sepal width for sepal length {new_sample.values.tolist()} is {predicted_width[0]:.2f} cm')

```

Output:

